

Florentin Coeurdoux

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WORK EXPERIENCE

Capital Fund Management (CFM)

Paris, France

VP Research Scientist

Jan 2024 – Present

- Lead a quantitative research team building and deploying machine learning models for live investment strategies, from research to production.
- Conduct foundational research on generative models as part of the CFM AI Lab, in collaboration with E. Vanden-Eijnden (Courant/NYU), S. Mallat (ENS/Flatiron Institute), and J.-P. Bouchaud (CFM); published 6 papers in 2025–2026 including at NeurIPS 2025 and ICML 2026.
- Supervise PhD students and research interns on topics at the intersection of generative modeling and statistical inference.

University of Oxford, Department of Statistics

Oxford, UK

Visiting Research Scholar

Oct 2022 – Mar 2023

- Conducted research on Monte Carlo methods and deep generative models under the supervision of Prof. Arnaud Doucet.

Credit Mutuel Arkéa

Rennes, France

Quantitative Developer

Apr 2019 – Nov 2020

- Developed a multivariate time series forecasting model for overdraft risk, deployed in production for 1.2 million users.
- Migrated option pricing codebase from VBA to C++, reducing execution time by 25% via importance sampling.

EDUCATION

Institut National Polytechnique (INP) – ANITI

Toulouse, France

Ph.D. in Applied Mathematics and Statistics

Nov 2020 – Dec 2023

- Supervised by Prof. Nicolas Dobigeon (IRIT/INP) and Prof. Pierre Chainais (Centrale Lille). Thesis on Monte Carlo sampling and deep generative models for Bayesian inference.

PUBLICATIONS

- 📄 E. Moreau, **F. Coeurdoux**, G. Ferré, E. Vanden-Eijnden, “Probing the Geometry of Diffusion Models with the String Method”, *International Conference on Machine Learning (ICML)*, Vancouver, 2026.
- 📄 H. Negrel, **F. Coeurdoux**, M. Albergo, E. Vanden-Eijnden, “Multitask Learning with Stochastic Interpolants”, *Neural Information Processing Systems (NeurIPS)*, 2025.
- 📄 **F. Coeurdoux**, E. Lempereur, N. Cuvelle-Magar, S. Mallat, E. Vanden-Eijnden, “Generative Modeling via Kernelized Stochastic Interpolants”, *ICML Workshop*, 2026.
- 📄 E. Lempereur, N. Cuvelle-Magar, **F. Coeurdoux**, S. Mallat, E. Vanden-Eijnden, “MGD: Moment Guided Diffusion for Maximum Entropy Generation”, *Mathematical Foundations of Machine Learning (MFML)*, 2026.
- 📄 M. Chalvidal, **F. Coeurdoux**, E. Vanden-Eijnden, “Covariance Shrinkage via Stochastic Interpolation”, *Preprint*, 2026.
- 📄 **F. Coeurdoux**, G. Ferré, J.-P. Bouchaud, “Random Matrix Theory of Early-Stopped Gradient Flow”, *Preprint*, 2026.
- 📄 **F. Coeurdoux**, N. Dobigeon, P. Chainais, “Plug-and-Play Split Gibbs Sampler: Embedding Deep Generative Priors in Bayesian Inference”, *IEEE Transactions on Image Processing*, 2024.

- 📖 **F. Coeurdoux**, N. Dobigeon, P. Chainais, “Normalizing Flow Sampling with Langevin Dynamics in the Latent Space”, *Machine Learning*, 2023.
- 📖 **F. Coeurdoux**, N. Dobigeon, P. Chainais, “Learning Optimal Transport Between Two Empirical Distributions with Normalizing Flows”, *European Conference on Machine Learning (ECML-PKDD)*, Grenoble, 2022.
- 📖 **F. Coeurdoux**, N. Dobigeon, P. Chainais, “Sliced-Wasserstein Normalizing Flows: Beyond Maximum Likelihood Training”, *European Symposium on Artificial Neural Networks (ESANN)*, Bruges, 2022.

INVITED TALKS

- 👥 Workshop: *Geostat Days*, “Solving Inverse Problems with Deep Learning”, Mines Paris PSL, Sept 2023.
- 👥 Seminar: *SIOP Seminar*, “Plug-and-Play Split Gibbs Sampler”, University of Bordeaux, May 2023.
- 👥 Workshop: *Interfacing Bayesian Statistics and Machine Learning*, “Langevin-based Normalizing Flow Sampling”, Bayes Centre, Edinburgh, Jan 2023.
- 👥 Seminar: *D^2 Reading Group*, “Normalizing Flow Sampling with Langevin Dynamics”, University of Oxford, Dec 2022.

SERVICE

Reviewer: NeurIPS, ICML, ICLR, AISTATS.

SKILLS

Programming: Python, C++, R, $\text{\LaTeX}$.

ML Frameworks: PyTorch, JAX.

Technologies: Docker, Git, Linux, AWS, SQL, Distributed Systems.

Languages: French (Native), English (Fluent).